Representation, Stability, and Comparative Behavior in Hybrid Quantum-Classical Machine Learning

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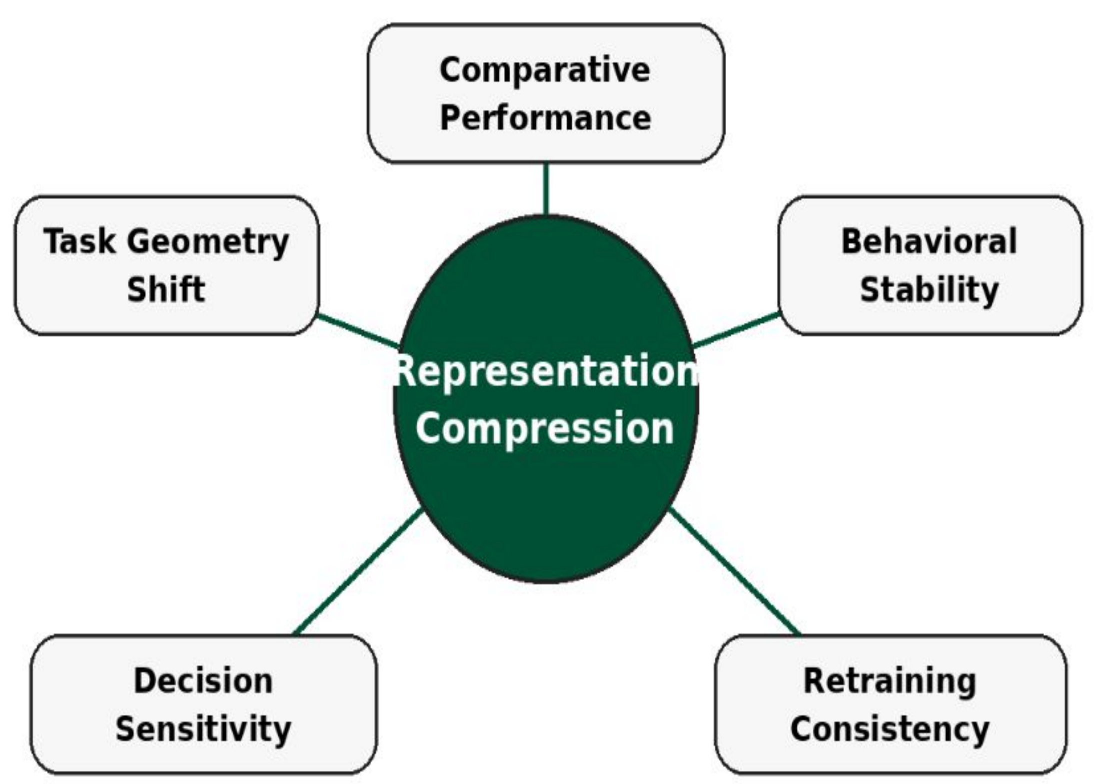
Introduction

- Hybrid quantum-classical models combine conventional machine learning with quantum circuits, which may cause them to respond differently to changes in input structure than classical models alone.
- This work tests whether representational compression does more than simplify the data.
- Specifically, I ask whether compression changes the structure of the learning problem itself, altering model behavior, comparative performance, and stability across retraining.ry of the input space is changed.

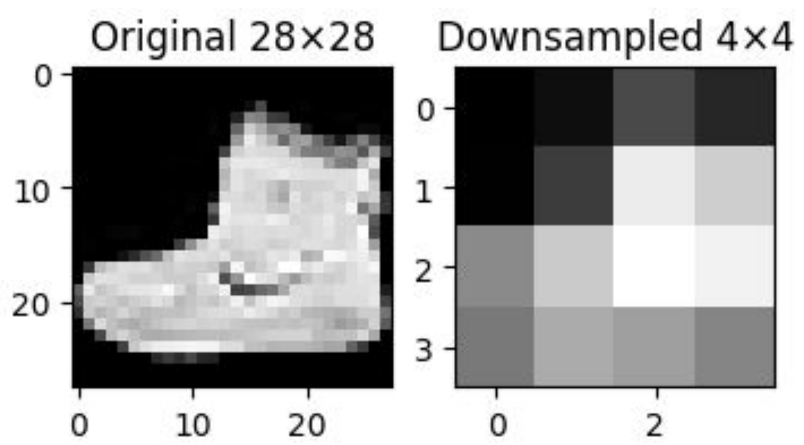
Objectives



How does compression reshape the learning problem?

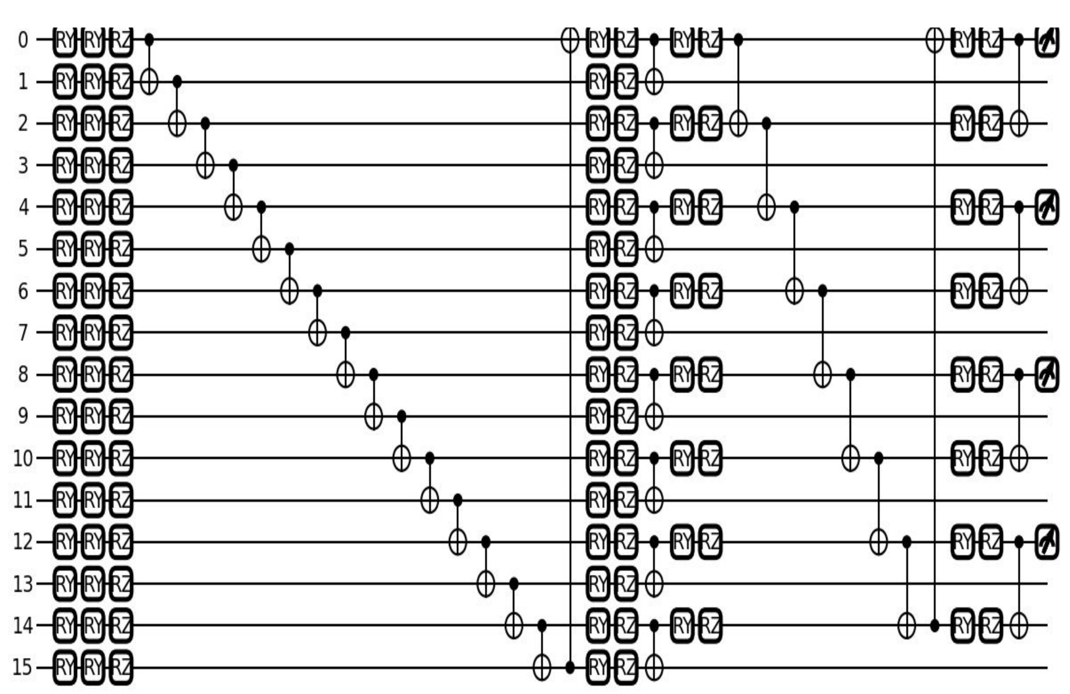
Method

- Matched hybrid quantum-classical and classical models were evaluated on the same image classification task under both original and compressed input conditions.
- Compression was treated not only as a reduction in detail, but as a change in how the learning problem is presented to the model.
- Comparative analysis examined performance, retraining stability, internal drift, and sensitivity to controlled perturbations.

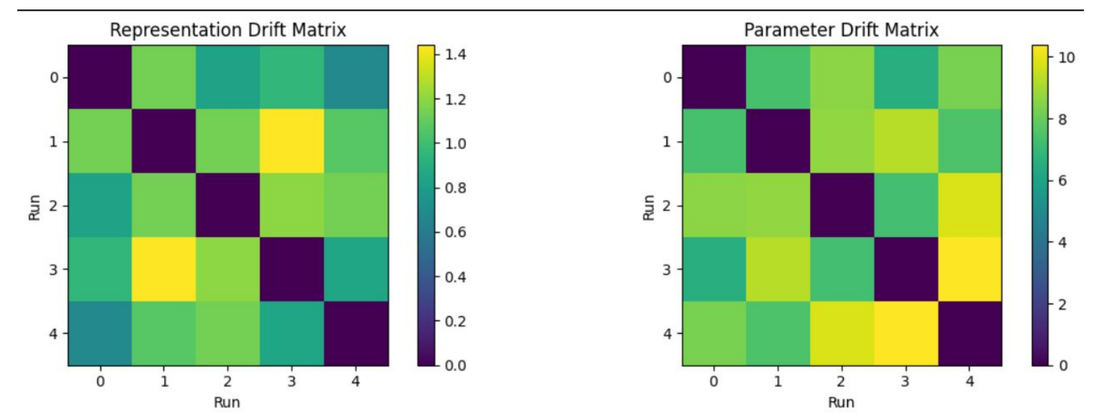


Evaluation Measures

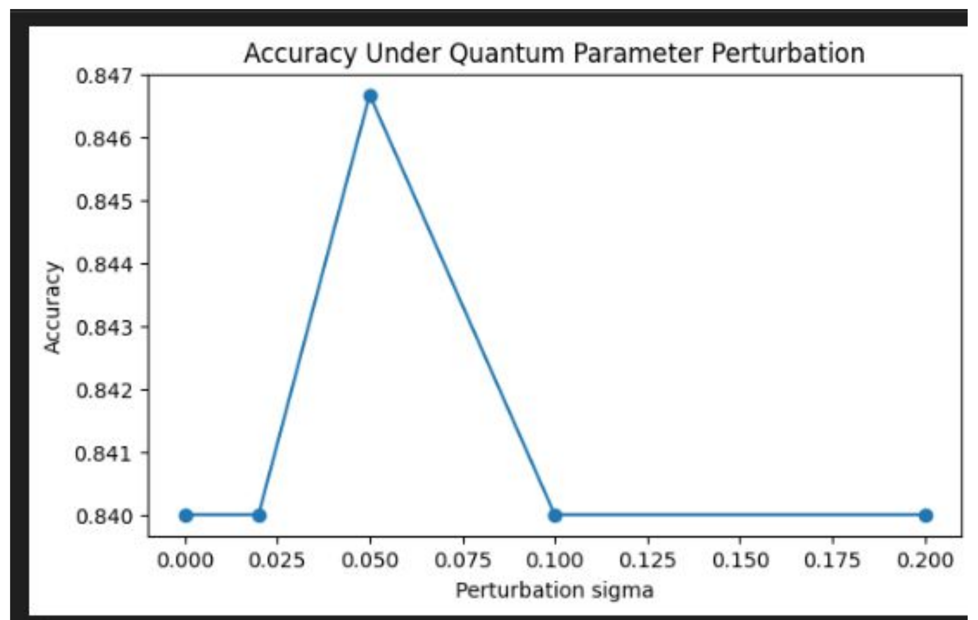
- Predictive performance: training, validation, and test accuracy across input regimes
- Optimization behavior: loss trajectories across epochs
- Behavioral stability: prediction consistency across repeated training runs
- Drift and sensitivity: representation drift, parameter drift, and response to controlled perturbation

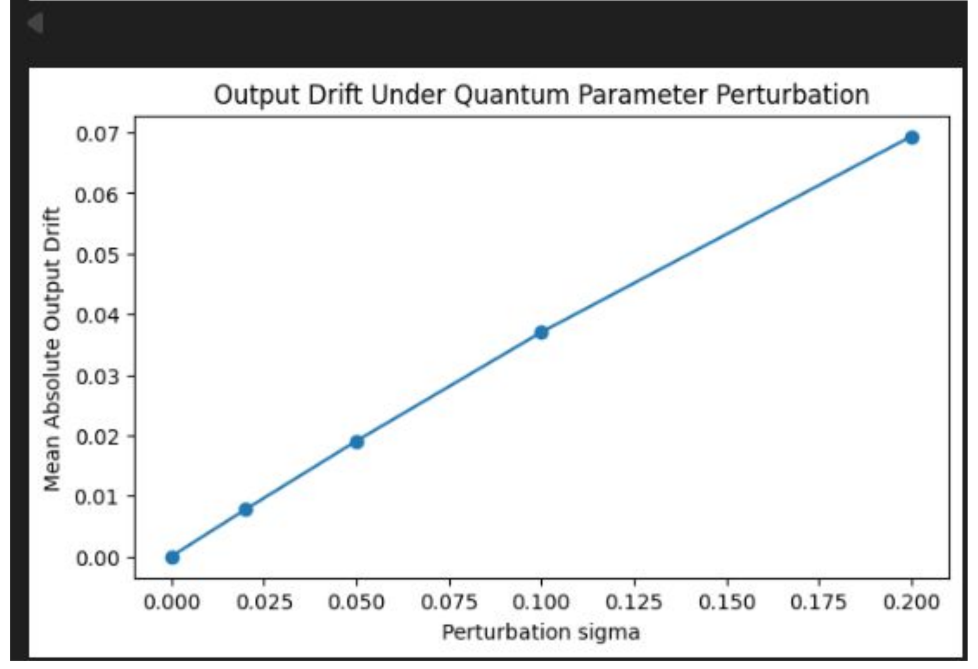


Results

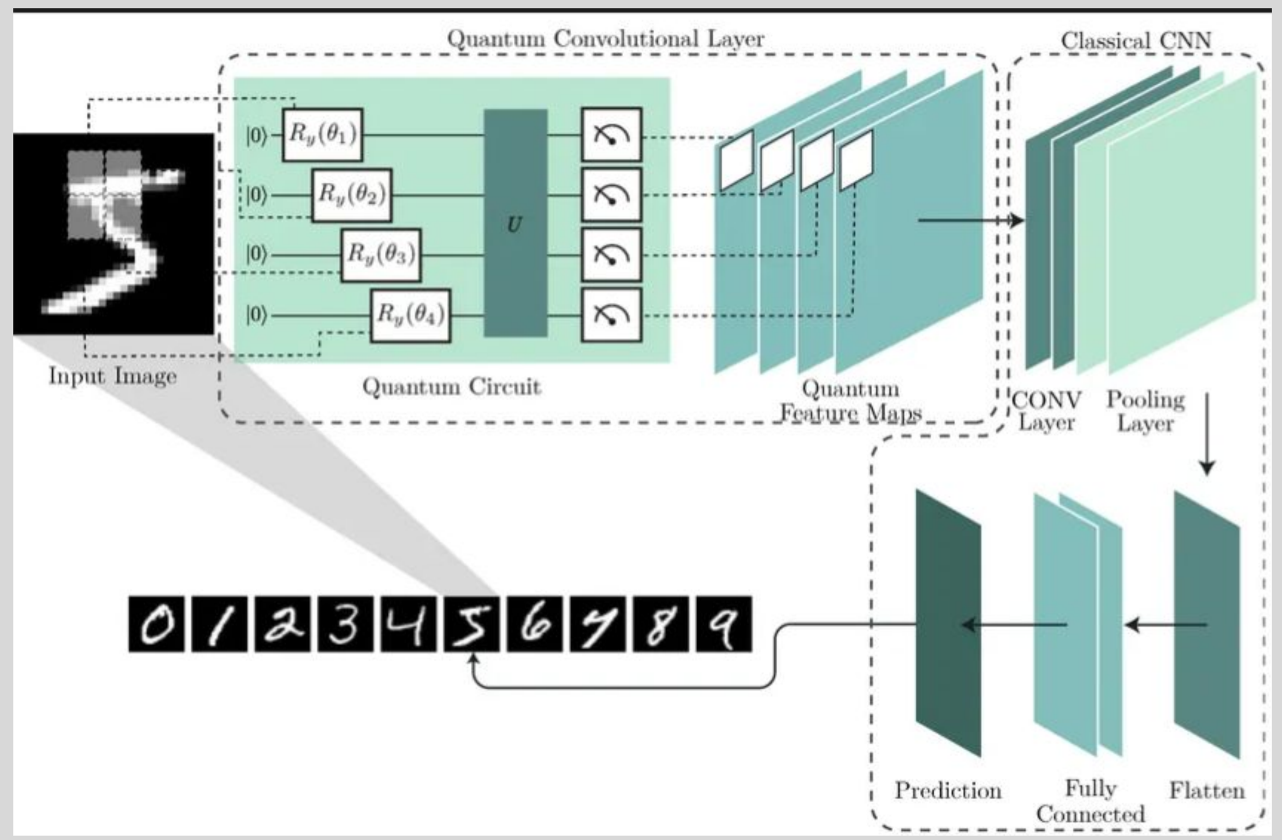


Pairwise representation and parameter drift across repeated training runs of the hybrid model. The persistence of nontrivial distances across runs suggests that retraining leads to distinct internal solutions, indicating instability at both the representational and parameter levels.





Sensitivity of the hybrid model to quantum parameter perturbation. While aggregate accuracy remains largely unchanged across modest perturbation scales, output drift rises monotonically, suggesting that behavioral sensitivity persists even when benchmark performance appears stable.



Conclusions

- Representational compression may change the learning problem itself, not merely reduce input detail.
- Because similar accuracy can mask differences in internal behavior, hybrid quantum-classical models should be evaluated using behavioral criteria in addition to aggregate performance.
- This is especially relevant for domains such as healthcare, where data are often noisy, incomplete, or heavily processed, and where model reliability, consistency, and robustness are critical.
- This perspective may help guide the development of more reliable hybrid learning systems for high-stakes settings including healthcare, scientific imaging, and decision-support applications.

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